So, $I.F. = xy^2$

So, $M' = 2xy^3 + 6x^2y^4$, $N' = 3x^2y^2 + 8x^3y^3$

So, $\int_{y=\text{const.}} M'\,dx = \int (2xy^3 + 6x^2y^4)\,dx = x^2y^3 + 2x^3y^4$.

and $\int_{x=\text{const.}} N'\,dy = \int (3x^2y^2 + 8x^3y^3)\,dy = x^2y^3 + 2x^3y^4$

Hence, the required general solution is

$$\boxed{x^2y^3 + 2x^3y^4 = c}$$ Ans.//

Differential equation of the first order and higher degree:

So far (till now), we have discussed differential equation of the first order and first degree. Now, we shall study, D.E. of first order and degree higher than the first (one).

For convenience, we denote $\frac{dy}{dx} = p$

The general form of a D.E. of first order and n^{th} degree is of the form

$$p^n + P_1 p^{n-1} + P_2 p^{n-2} + \cdots + P_n = 0 \quad \text{——} \quad (1.)$$

where $P_1, P_2, \ldots P_n$ are functions of x and y.

We can solve D.E. (1.) by three ways

(i.) Equations solvable for p

(ii.) Equations solvable for y, $[y = f(x,p)]$, (useful when x is absent).

(iii.) Equations solvable for x, $[x = f(y,p)]$, (useful when y is absent.)

Equations Solvable for p :→ If possible, resolve the left hand side of (1) into n - linear factors, like $[p - f_1(x,y)][p - f_2(x,y)] \cdots [p - f_n(x,y)] = 0$.

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which is equivalent to

$$p - f_1(x,y) = 0 \quad , \quad p - f_2(x,y) = 0 \quad , \quad \cdots \quad p - f_n(x,y) = 0.$$

Each of these equations is of the first order and first degree and can be solved by the method already discussed.

→ Suppose solutions are

$$f_1(x,y,c) = 0 \quad , \quad f_2(x,y,c) = 0 \quad , \quad \cdots \quad , \quad f_n(x,y,c) = 0$$

On combining these into one equation, we get the solution of (1) as

$$\boxed{f_1(x,y,c) \cdot f_2(x,y,c) \cdot \cdots \cdot f_n(x,y,c) = 0}$$

Ex.b. Solve $p^2 + p - 6 = 0$,

$$\boxed{p \equiv \frac{dy}{dx}}$$

Solution.

$$p^2 + 3p - 2p - 6 = 0$$
$$p(p+3) - 2(p+3) = 0$$
$$(p+3)(p-2) = 0$$

So, $p + 3 = 0$

$$\frac{dy}{dx} + 3 = 0$$
$$dy + 3dx = 0$$
$$\boxed{y + 3x - c = 0}$$

$p - 2 = 0$

$$\frac{dy}{dx} - 2 = 0$$
$$dy - 2dx = 0$$
$$\boxed{y - 2x - c = 0}$$

Combining these two, we get

$$\boxed{(y + 3x - c)(y - 2x - c) = 0}$$

which is the general solution of the given D.E.

Ex.b. Solve. $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$

Sol^n. On factorizing, we get $(yp + 3x)(xp - 2y) = 0$

$yp + 3x = 0$

$$y\frac{dy}{dx} + 3x = 0$$
$$y\,dy + 3x\,dx = 0$$
$$y^2 + 3x^2 = c$$
$$\boxed{y^2 + 3x^2 - c = 0}$$

$xp - 2y = 0$

$$x\frac{dy}{dx} - 2y = 0$$
$$\frac{dy}{y} = \frac{2\,dx}{x}$$
$$\log y = 2\log x + \log c$$
$$y = cx^2$$
$$\text{or } \boxed{y - cx^2 = 0}$$

On combining these two, we get

$$(y^2 + 3x^2 - c)(y - cx^2) = 0$$

which is the general solution of the given Differential equation.

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Exp: Solve $x^2\left(\frac{dy}{dx}\right)^2 + xy\frac{dy}{dx} - 6y^2 = 0$

Sol^n: The given equation is $x^2p^2 + xyp - 6y^2 = 0$

On factorization, we get

$$x^2p^2 + 3xyp - 2xyp - 6y^2 = 0$$
$$xp(xp + 3y) - 2y(xp + 3y) = 0$$
$$(xp + 3y)(xp - 2y) = 0$$

Then

$xp + 3y = 0$

$x\frac{dy}{dx} + 3y = 0$

$x\frac{dy}{dx} = -3y$

$\frac{dy}{y} + 3\frac{dx}{x} = 0$

$\log y + 3\log x = \log c$

$\log y + \log x^3 = \log c$

or $yx^3 = c$

or $\boxed{yx^3 - c = 0}$

$xp - 2y = 0$

$x\frac{dy}{dx} - 2y = 0$

$\frac{dy}{y} - 2\frac{dx}{x} = 0$

$\log y - 2\log x = \log c$

$\log y = \log cx^2$

or $y = cx^2$

or $\boxed{y - cx^2 = 0}$

Hence, the general solution is

$$\boxed{(yx^3 - c)(y - cx^2) = 0}$$

(*) Equations Solvable for y. :-

This method is applicable if the D.E. of first order and first degree can be written in the form

$$\boxed{y = f(x, p)} \quad \text{---(1)}$$

i.e., when the given D.E. is of the first degree in y, then we can apply this method

On differentiating eqn (1), w.r.to x, we get a relation of the form

$$\frac{dy}{dx} = \phi\left(x, p, \frac{dp}{dx}\right)$$

i.e. $$\boxed{p = \phi\left(x, p, \frac{dp}{dx}\right)} \quad \text{---(2)}$$

which is a differential eqn of first order and first degree in p and x. It can be solved using the previous method and suppose its solution is $\boxed{F(x, p, c) = 0}$ ---(3).

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Then, on eliminating p from (1) and (2), we get the required solution.

Note 1: If p can not be eliminated easily, then we say that equation (1) and (2) constitute the general solution.

Note 2: → This method is especially useful for the equations in which x is absent.

Exp. Solve $y + xp = x^4 p^2$

Soln. Solving for y, put the differential equation in the form

$$y = -xp + x^4 p^2 \quad \text{——— (1)}$$

Differentiating with respect to x, we get

$$p = -p - x\frac{dp}{dx} + 4x^3 p^2 + 2x^4 p \frac{dp}{dx}$$

$$2p - 4p^2 x^3 + (x - 2x^4 p)\frac{dp}{dx} = 0$$

$$2p(1 - 2px^3) + x\frac{dp}{dx}(1 - 2x^3 p) = 0$$

$$(1 - 2px^3)\left(2p + x\frac{dp}{dx}\right) = 0$$

gives Singular Soln.

for the general solution, we have $2p + x\frac{dp}{dx} = 0$

$$x\frac{dp}{dx} = -2p$$

$$\frac{dp}{p} = -\frac{2dx}{x}$$

$$\log p = -\log x^2 + \log c$$

$$\text{or} \quad p = \frac{c}{x^2}$$

Now, eliminate p between (1) and (2), we get or $\boxed{px^2 = c}$ or $\boxed{p = \frac{c}{x^2}}$ → (2)

$$y = -x\frac{c}{x^2} + x^4\frac{c^2}{x^4}$$

$$\text{or} \quad y = -\frac{c}{x} + c^2$$

$$\text{or} \quad \boxed{xy + c = c^2 x}$$ → This is the general solution of the given D.E.

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Exp. Solve $3p^5 - py + 1 = 0$

Soln → Solving for y, $y = 3p^4 + p^{-1}$ ——— (1.)

Diff. (1) w.r.to x, we get

$$p = 12p^3 \frac{dp}{dx} - \frac{1}{p^2}\frac{dp}{dx}$$

$$\left(p - 12p^3 \frac{dp}{dx} + p^{-2}\frac{dp}{dx}\right) = 0$$

$$p\left(1 - 12p^2 \frac{dp}{dx} + p^{-3}\frac{dp}{dx}\right) = 0$$

either $p = 0$ gives singular soln

or $1 - 12p^2\frac{dp}{dx} - p^{-3}\frac{dp}{dx} = 0$

$$1 - (12p^2 - p^{-3})\frac{dp}{dx} = 0$$

$$dx = (12p^2 - p^{-3})\,dp$$

$$\boxed{x = 4p^3 + \frac{1}{2}p^{-2} + c} \quad \text{——— (2.)}$$

So, the original equation, $y = 3p^4 + p^{-1}$

and the equation $x = 4p^3 + \frac{1}{2}p^{-2} + c$

constitute the general solution.

OR you can write, by eliminating p from (1) and (2), we get the required general solution).

(*) Equations Solvable for x →

When the given Differential equation is of first degree in x, then it can be put in the form

$$x = f(y, p) \quad \text{——— (1.)}$$

and can be solved for x.

Diff eqn (1), w.r.to y, we get an equation of the form

$$\frac{dx}{dy} = F\left(y, p, \frac{dp}{dy}\right) \quad \text{———}$$

or $$\frac{1}{p} = F\left(y, p, \frac{dp}{dy}\right) \quad \text{——— (2.)}$$

which is a Differential equation in two variables p and y, which can be easily solved. Suppose, the solution of (2) is $\phi(y, p, c) = 0$ ——— (3.)

Then, on eliminating p from (1) and (2), we get the required solution.

Note 1 → If p can not be eliminated easily, then we say that eqn (1) and (3) constitute the general solution.

Note 2 → This method is especially useful for eqtions in which y is absent.

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(28) Exp: Solve $y = 2px + y^2p^3$

Solution :- Solving for x, write the D.E. in the following form

$$2px = y - y^2p^3$$

$$\text{or } x = \frac{1}{2}\left(\frac{y}{p} - y^2p^2\right) \quad \text{---------- (1.)}$$

Differentiating both sides with respect to y, we get

$$\frac{dx}{dy} = \frac{1}{p} = \frac{1}{2}\left(\frac{1}{p} - \frac{y}{p^2}\frac{dp}{dy} - 2yp^2 - 2y^2p\frac{dp}{dy}\right)$$

$$\text{or } 2p = p - y\frac{dp}{dy} - 2yp^4 - 2y^2p^3\frac{dp}{dy}$$

$$p + 2yp^4 + y\frac{dp}{dy} + 2y^2p^3\frac{dp}{dy} = 0$$

$$p(1 + 2yp^3) + y\frac{dp}{dy}(1 + 2yp^3) = 0$$

$$\left(p + y\frac{dp}{dy}\right)(1 + 2yp^3) = 0$$

Discard the factor $1 + 2yp^3 = 0$, we have.

(It gives singular solution)

$$p + y\frac{dp}{dy} = 0$$

$$\text{or } \frac{dy}{y} + \frac{dp}{p} = 0$$

$$\log y + \log p = c$$

$$yp = c$$

$$\text{or } \boxed{p = c/y} \quad \text{---------- (2)}$$

By putting this value of p in the given equation, we have

$$y = \frac{2cx}{y} + \frac{c^3}{y}$$

$$\text{or } \boxed{y^2 = 2cx + c^3}$$ Ans.//

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